

Section I

10 marks

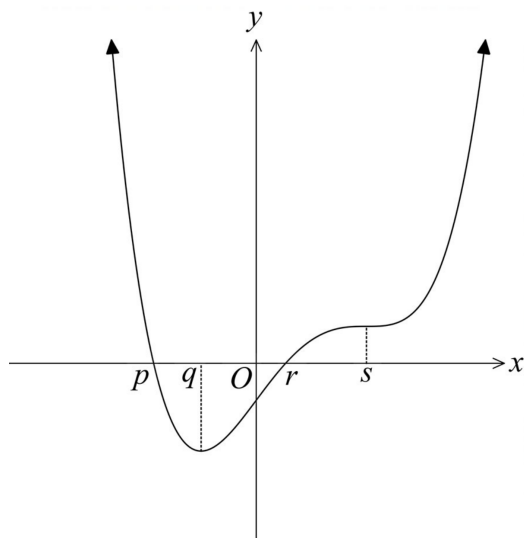
Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 The graph of e^{-x} is:
- (A) Monotonically increasing
 - (B) Monotonically decreasing
 - (C) Neither monotonically increasing nor monotonically decreasing
 - (D) Cannot be determined
- 2 What are the coordinates of the turning point to the curve $y = e^x - ex$?
- (A) $(1, 0)$
 - (B) $(0, 1)$
 - (C) $(1, e)$
 - (D) $(e, 1)$
- 3 The derivative of $(x^3 + \ln 3)^3$ is:
- (A) $3 \times (x^3 + \ln 3)^2$
 - (B) $3 \times 3x^2 \times \frac{1}{3}(x^3 + \ln 3)^2$
 - (C) $3 \times 3x^2 \times (x^3 + \ln 3)^2$
 - (D) $3 \times \left(3x^2 + \frac{1}{3}\right)(x^3 + \ln 3)^2$
- 4 The values of the solutions to $|x + 4| = |2x + 5|$ are:
- (A) $x = 1$ or 3
 - (B) $x = -1$ or -3
 - (C) $x = -1$ or 3
 - (D) $x = 1$ or -3
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- 5 The diagram shows the graph of $f'(x)$ for the function $f(x)$.



For what value of x does the function $f(x)$ have a maximum turning point?

- (A) $x = p$
 - (B) $x = q$
 - (C) $x = r$
 - (D) $x = s$
- 6 The primitive function of $(3x - 2)^3$ is:

- (A) $9(3x - 2)^2$
- (B) $\frac{(3x - 2)^4}{12}$
- (C) $\frac{(3x - 2)^4}{4}$
- (D) $9(3x - 2)^4$

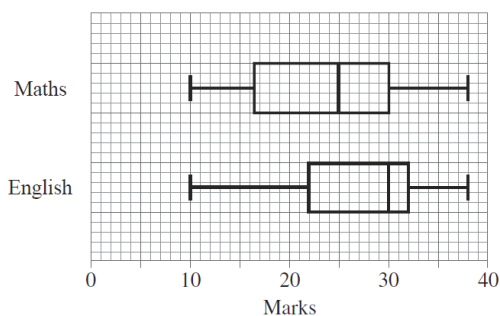
- 7 Jack purchases a new car for \$12,600 and agrees to pay over monthly instalments over 2 years. The car dealer charges reducible interest of 8.2%pa with monthly repayments of \$620. The equivalent flat interest rate is:

- (A) 4.9%pa
- (B) 8.2%pa
- (C) 9.0%pa
- (D) 17.9%pa

- 8 If $\log_{10} 19 = a$, then $\log_{10} \left(\frac{1}{190} \right)$ is equal to:

- (A) $-(1 + a)$
- (B) $(1 + a)^{-1}$
- (C) $\frac{a}{10}$
- (D) $\frac{1}{10a}$

- 9 The parallel-box plot show the results of a Maths exam and an English exam.



In Maths, 112 students completed the test. The number of students who scored above 30 marks was the same for the Maths exam and the English exam.

How many students completed the English exam?

- (A) 8
- (B) 50
- (C) 56
- (D) 112

- 10** A probability density function is defined as $y = 2x$ between $0 \leq x \leq 1$. The sum of the mean and the variance is:

- (A) $\frac{2}{3}$
 - (B) $\frac{13}{18}$
 - (C) $\frac{4}{9}$
 - (D) $\frac{1}{18}$
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Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

Find the equation of the normal to the curve $xy = -4$ at the point $(-1, 4)$. 2

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Question 12 (3 marks)

Solve $\tan \theta = \frac{1}{\sqrt{2}}$, for $-\pi \leq \theta \leq \pi$, correct to two significant figures. 3

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Question 13 (2 marks)

Show that $a^{\log_b x} = x^{\log_b a}$.

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Question 14 (2 marks)

Each week, person A and person B enter different raffles. The probability that person A wins is $\frac{1}{16}$. **2**
The probability that person B wins is $\frac{1}{9}$. What is the probability that during the next three weeks, at least one of them wins a prize?

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Question 15 (7 marks)

A function is defined by $f(x) = (x + 2)(x^2 - 3x + 5)$.

- a) Find the stationary point/s and determine their nature. **3**

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- b) Show that $f(x) \geq 0$ for $x \geq -2$. **2**

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Question 16 (1 marks)Differentiate $\log_2 x$.**1**

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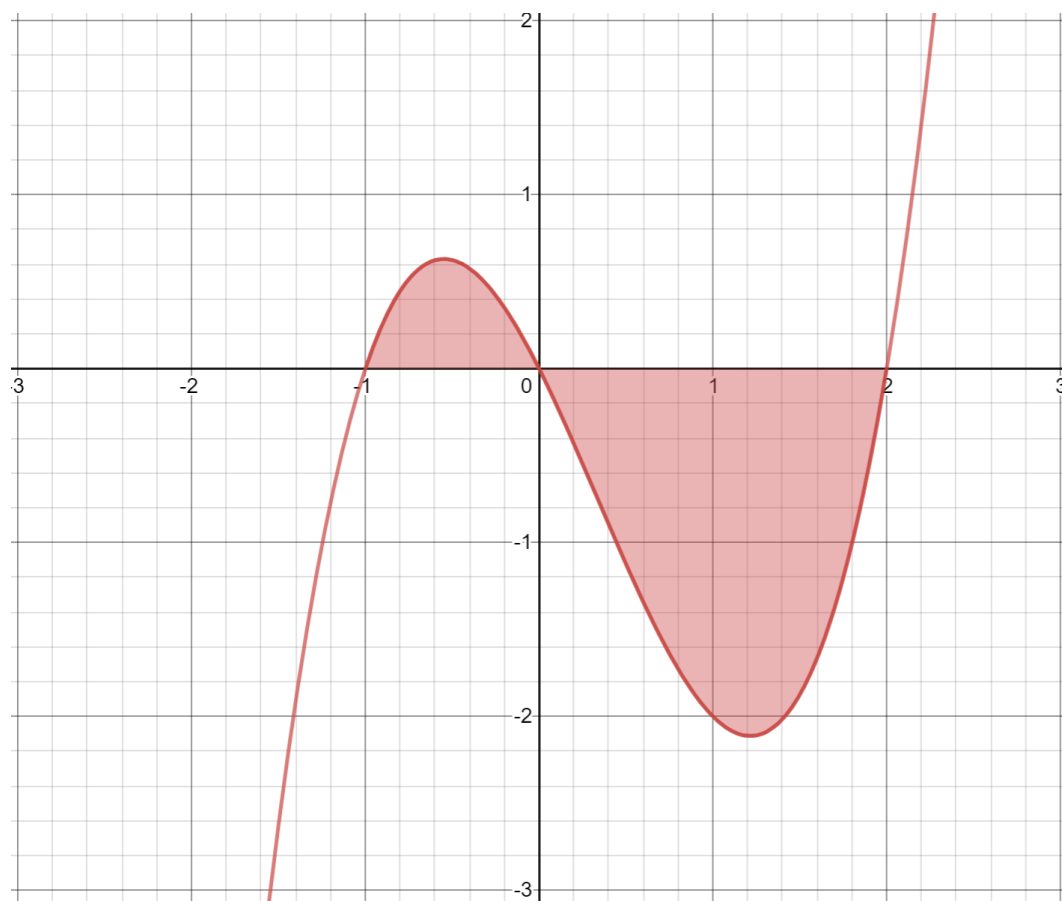
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Question 17 (2 marks)Differentiate $\tan(e^x)$.**2**

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Question 18 (6 marks)The diagram shows the curve $y = x^3 - x^2 - 2x$.

a) Find the area of the shaded region.

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b) Use the trapezoidal rule and four function values to approximate the shaded region between $0 \leq x \leq 2$.

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c) Find the percentage error of the approximation above.

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Question 19 (3 marks)

A discrete random variable X has the probability distribution table shown.

$X = x$	17	18	19	20
$P(x)$	0.1	0.4	0.3	?

Calculate the variance of X .

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Question 20 (3 marks)

Let $f(x) = A \sin x + B \cos x$, where A and B are constants. Find the value of A and B if $f\left(\frac{\pi}{4}\right) = 3$ **3**
and $f'\left(\frac{\pi}{4}\right) = 1$.

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Question 21 (3 marks)

Using the fact that $\sin^2 x + \cos^2 x = 1$, prove that $1 + \tan^2 x = \sec^2 x$ and $1 + \cot^2 x = \csc^2 x$. **3**

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Question 23 (3 marks)

A function is transformed from $f(x) = \sin x$ to $g(x) = \cos x + 1$ through horizontal and vertical shift. **3**
Describe the transformations occurred, expressing $\cos x + 1$ in terms of $f(x)$.

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Question 24 (3 marks)

Find the range of $x^2 + y^2 + 2x + 6y = -4$ in interval notation. **3**

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Question 25 (4 marks)

A school’s soccer captain notices that the team has a probability of 60% of winning the game if it rains, but only 25% if it is dry. The weather station reports a 30% chance that it rains on the day of the match.

- a) Find the probability that they do not win. **2**

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- b) Given that the team has won the game, find the probability that it rained on the day of the match. **2**

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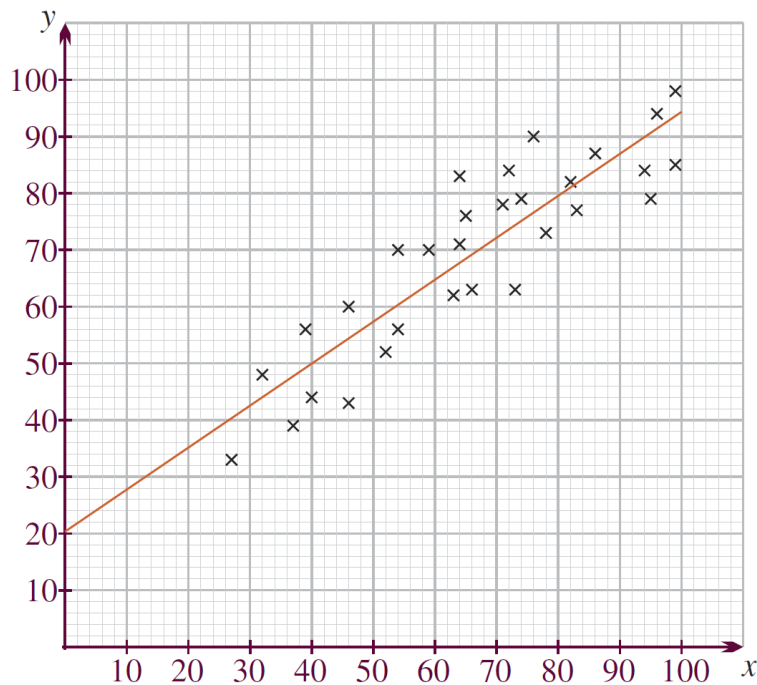
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Question 26 (3 marks)

The following scatterplot shows the relationship between internal marks (x) and external marks (y) for the HSC course Mathematics Advanced at a high school. A line of best fit was drawn.



- a) Calculate the equation of the line of best fit.

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- b) NESAs was unable to obtain the internal marks for a student, but the student had obtained 30 for his external marks. Calculate his predicted internal marks based off your answer in part a).

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Question 27 (6 marks)

A cylindrical can with height h and radius r with volume 400cm^3 .

- a) Show that the surface area is $A = \frac{800}{r} + 2\pi r^2$. **2**

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- b) Hence, find the radius, in exact form that minimises the surface area. **4**

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Question 28 (4 marks)

A sheet of paper is 0.05mm thick.

- a) The paper is folded three times, doubling the thickness with each fold. Find the thickness of the paper after these folds. **1**

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- b) After how many folds does the paper exceed a thickness of 50cm? **3**

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Question 29 (7 marks)

A probability density function is used to model the reaction time of individuals, defined by

$$f(t) = \begin{cases} k(1 - t^2) & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- a) Find the value of k . **2**

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b) Find the mean and the variance of the probability density function. **3**

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c) Sam has a reaction time of 0.4 seconds. Find his z -score. **2**

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Question 31 (5 marks)

Player A and B compete in a series of tennis matches. Player A has a probability p of winning a match while Player B has a probability q of winning a match. To win the series, a player must win two matches in a row.

- a) If player A wins the first match, show that the probability that Player A wins is $\frac{p^2}{1 - pq}$. **3**

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- b) Hence, by considering if Player B wins the first match, show that the probability that Player A **2**
wins is $\frac{p^2(1 + q)}{1 - pq}$.

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Question 32 (6 marks)

Two functions are defined by $f(x) = x - 10 + \frac{25}{x}$ and $g(x) = x - 8 + \frac{15}{x}$.

- a) Find a simplified expression for $\frac{f(x)}{g(x)}$. **2**

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- b) Find the domain for $\frac{f(x)}{g(x)}$ in interval notation. **2**

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Question 33 (6 marks)

- a) Show that the limiting sum of $1 + 4 \sin^2 x \cos^2 x + 16 \sin^4 x \cos^4 x + 64 \sin^6 x \cos^6 x \cdots$ is $\sec^2 2x$. **3**

- b) Hence, evaluate $\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} 1 + 4 \sin^2 x \cos^2 x + 16 \sin^4 x \cos^4 x + 64 \sin^6 x \cos^6 x \cdots dx$. **3**